

KSHITIJ DURAPHE

Boston, MA | kshitijduraphe5@gmail.com | +1 (314) 886-3066 | ksd3.github.io | github.com/ksd3 | [LinkedIn](#)

Machine learning engineer and researcher working on foundation models, evaluation, and scientific inverse problems. Co-first author on a **NeurIPS ML4PS 2025 Spotlight** (top 1%); first author in *The Astrophysical Journal*. Builds the infrastructure and the measurements that show whether models actually work.

EXPERIENCE

Thespian Labs

Somerville, MA

AI Engineer

Nov 2025 – Feb 2026

- Trained **Text2Motion** foundation models with **VQ-VAE** tokenization over 5000+ hours of human-performance time-series; implemented DARTControl-style **RL post-training** with reward modeling and rollout infrastructure on GCP.

Absentia Technologies

Boston, MA

Founding Machine Learning Engineer

Jan 2025 – Nov 2025

- Architected a production SaaS video-analysis platform on **AWS** from scratch — core AI service infrastructure plus Terraform/Docker CI/CD, cutting deployment cycles from days to **<2 h**.
- Built the APIs and infrastructure for autonomous agents to read video streams, reason over events, and flag anomalies in real time (Python, PyTorch, Kafka, Kubernetes).
- Established an automated **evaluation pipeline** benchmarking agent accuracy and API latency; reduced regressions **40%** and raised stability to **99.9%**, including resolving a production memory leak on call.
- Implemented **distributed training** with PyTorch FSDP (sharding, mixed precision, checkpointing, fault recovery); built a diffusion-based synthetic-data pipeline that cut false positives **15%**.

Halo AI (Columbia-incubated stealth)

New York, NY (Remote)

Founding AI Engineer

Dec 2023 – Aug 2024

- Built a federated-learning pipeline and LLM ensemble for on-device agentic assistants, cutting inference latency **27%** to **<1.5 s**.
- Architected an AWS MLOps pipeline (EC2, S3, Lambda, SageMaker) automating CI/CD and evaluation; reduced data-preparation time **70%**. Applied distillation and 8-bit quantization for **80%** faster inference at **90%** accuracy retention.

The KeelWorks Foundation

Oak Harbor, WA (Remote)

Software Engineer, ML Applications

Jul 2024 – Jan 2025

- Engineered a production **RAG** retrieval-and-reasoning system (TypeScript/Python, LangChain) serving a search API at **<5 s p95** with **92% context relevance (MRR)** over 2,500+ documents on PostgreSQL.
- Reduced model size **50%** and improved inference speed **80%** via **8-bit GPTQ quantization** and distillation while holding **>90%** task accuracy.

RESEARCH

University of Zielona Góra

Zielona Góra, Poland (Remote)

Research Assistant

Mar 2025 – Present

Advisor: Prof. [Gopal Bhatta](#)

- First-authored a 12-year NuSTAR archival timing study of Cygnus X-1 (*ApJ*, 2026); reduced and analyzed the complete public archive, led manuscript and referee response.

UniverseTBD Collaboration

Remote

Researcher

Jul 2024 – Present

Collaborators: Dr. [Michael J. Smith](#), Dr. [John F. Wu](#)

- Co-first author on *The Platonic Universe* — NeurIPS ML4PS 2025 **Spotlight** (top 1%). Designed the cross-model representational-alignment **evaluation framework** spanning SDSS, DESI and JWST under model and data scaling.
- Creator and maintainer of open-source [platonic-universe](#): survey-aware loaders, reproducible eval harness, configs, CI. Run on a 6,000 GPU-hour NCSA DeltaAI allocation.

Space Physics Lab, Boston University

Boston, MA

- Architected a distributed ingestion pipeline (Kafka, Dask, S3) cutting processing of 3 TB+ sensor datasets from >24 h to <3 h; built generative inpainting and super-resolution pipelines to reconstruct corrupted sensor data.
- Developed automated detection over noisy all-sky imagery by adapting faint-source astronomical algorithms and evaluating deep-learning models (ConvNeXt).

SELECTED PUBLICATIONS

1. State-Dependent X-ray Variability in Cygnus X-1: A 12-Year NuSTAR Timing Study of Accretion Flow Geometry
K. Duraphe, G. Bhatta, K. Mandar, C. Khanal, et al.
The Astrophysical Journal, 2026

2. Cross-Modal Attention is Overparameterized: The Sufficiency of Modality-Level Alignment
K. Duraphe, K. Mohamed, N. Morgan
Submitted to NeurIPS Main Track, 2026

3. The Platonic Universe: Do Foundation Models See the Same Sky?
K. Duraphe, M. J. Smith, J. F. Wu, S. Sourav (co-first author)
NeurIPS ML4PS Workshop, 2025 — Spotlight (top 1%) expanded follow-up submitted to NeurIPS 2026 main track

OPEN SOURCE AND PROJECTS

ArkOS — MIT SIPB

Jun 2025 – Present

- Open-source local-LLM agent platform with long-term memory. Lead DevOps and developer documentation (Mintlify): GitHub Actions CI/CD, contributor onboarding, frontend/backend development.

rinexpy

2023 – Present

- Python library for fast RINEX GNSS reading, merging, clock-jump and higher-order ionospheric correction, and format conversion; a superset of georinex.

TECHNICAL SKILLS

Languages:	Python (primary), C, C++, TypeScript, SQL (Postgres), MATLAB
ML / AI:	PyTorch, FSDP, distributed training, RL post-training, VQ-VAE, foundation models / VLMs, RAG, LLM agents, evaluation systems, synthetic data, quantization and distillation
Cloud and MLOps:	AWS (EC2, S3, Lambda, SageMaker, EKS), GCP, Docker, Kubernetes, Terraform, Kafka, Dask, Slurm, MLflow
Python tooling:	pyproject.toml, uv, ruff, pytest, pre-commit, Sphinx, GitHub Actions
Methods:	Bayesian inference (MCMC), time-series and spectral analysis, inverse problems, uncertainty quantification, numerical ODE/PDE solvers

EDUCATION

Boston University

Boston, MA

M.S., Electrical and Computer Engineering (with Thesis)

Sep 2022 – May 2024 · GPA: 3.8/4.0

College of Engineering Pune

Pune, India

B.Tech., Electrical Engineering (Minor: Computer Science)

Aug 2018 – June 2022 · GPA: 9.11/10.0 (Department Bronze Medalist)

AWARDS

- NeurIPS ML4PS 2025 — Spotlight Paper (top 1%)

2025
 - Outstanding Reviewer, ICML 2026 Mechanistic Interpretability Workshop

2026
 - 2nd / 100+ international teams, MIT iQuHack 2023 (IBM × Covalent Challenge)

2023